

● EASY

● PROBLEM-SOLVING AND DATA ANALYSIS

Problem-Solving and Data Analysis

17

9 questions · ~14 min

Q1

PROBLEM-SOLVING AND DATA ANALYSIS

The population density of Worthington is 290 people per square mile. Worthington has a population of 92,800 people. What is the area, in square miles, of Worthington?

- A. 102,400
- B. 93,090
- C. 320
- D. 32

The high temperature, in degrees Fahrenheit ($^{\circ}\text{F}$), in a certain city was recorded for each of 5 days. The data are shown below.

Day	1	2	3	4	5
High temperature ($^{\circ}\text{F}$)	81	80	81	81	82

Over this 5-day period, which of the following is NOT equal to 81°F ?

- A. Median of the high temperatures
- B. Mean of the high temperatures
- C. Mode of the high temperatures
- D. Range of the high temperatures

Q3

PROBLEM-SOLVING AND DATA ANALYSIS

The number of raccoons in a 131-square-mile area is estimated to be 2,358. What is the estimated population density, in raccoons per square mile, of this area?

- A. 18
- B. 131
- C. 149
- D. 2,376

Q4

PROBLEM-SOLVING AND DATA ANALYSIS

2, 9, 14, 23, 32

What is the mean of the data shown?

- A. 14
- B. 16
- C. 17
- D. 32

Q5

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A giant armadillo has a mass of 39 kilograms. What is the giant armadillo's mass in **grams**? (1 kilogram = 1,000 grams)

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q6

GRID-IN

PROBLEM-SOLVING AND DATA ANALYSIS

A participant in a bicycle race completes the race with an average speed of 24,816 yards per hour. What is this average speed, in miles per hour? 1 mile = 1,760 yards

STUDENT-PRODUCED RESPONSE

.	/	.	.
0	0	0	0
1	1	1	1
2	2	2	2
3	3	3	3
4	4	4	4
5	5	5	5
6	6	6	6
7	7	7	7
8	8	8	8
9	9	9	9

Write the answer above. Fill the matching bubble for each digit, plus the fraction bar or decimal point if needed.

Q7

PROBLEM-SOLVING AND DATA ANALYSIS

How many teaspoons are equivalent to 44 tablespoons? 3 teaspoons = 1 tablespoon

- A. 47
- B. 88
- C. 132
- D. 176

The ratio x to y is equivalent to the ratio 12 to t . When $x = 156$, what is the value of y in terms of t ?

- A. $13t$
- B. $12t$
- C. $144t$
- D. $168t$

Q9

PROBLEM-SOLVING AND DATA ANALYSIS

How many meters are equivalent to 2,300 centimeters? 100 centimeters = 1 meter

- A. 0.043
- B. 23
- C. 2,400
- D. 230,000